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Agentic artificial intelligence in 2024–2025: technological innovations and application potential in economic applications

Abstract:

In 2024–2025, the development of artificial intelligence entered a new stage in which agentic systems are becoming increasingly important, extending the capabilities of large language models by integrating mechanisms of autonomy, long-term memory, multi-stage planning, and interaction with the environment. AI agents are no longer merely tools that support users, but are increasingly acting as active participants in decision-making processes in organizations, public administration, and the digital economy sector. This phenomenon is directly linked to the dynamic development of generative AI, the increase in the computing power of data centers, and the growing scale of capital investment in cloud infrastructure and data ecosystems. From an economic perspective, agentic AI offers significant potential for productivity growth, automation of complex management processes, and optimization of resource allocation, while redefining labor market structures and value creation models. The article analyzes the interdisciplinary determinants of this development, combining IT, economic, legal, and ethical perspectives. Particular attention is paid to agent architectures, cause-and-effect reasoning capabilities, and Theory of Mind concepts in the context of LLMs and multi-agent systems. It is pointed out that the growing autonomy of agents increases both their usefulness and the level of operational risk, including the risk of erroneous decisions made without direct human control. An important theme of the analysis is cybersecurity issues, including the vulnerability of AI agents to manipulation, prompt injection attacks, abuse in autonomous systems, and threats to data integrity and confidentiality. The author also emphasizes the importance of economic security, especially in the context of technology concentration, asymmetry of access to advanced AI, and potential implications for the competitiveness of companies and countries. The ethical section discusses issues of responsibility for decisions made by autonomous systems, algorithmic transparency, and the compliance of AI agents with social values. The overall conclusion is that the development of agentic AI should be geared towards increasing efficiency and

innovation while strengthening mechanisms for oversight, security, and responsible implementation of the technology.

Keywords:

agentic artificial intelligence, autonomous systems, large language models, generative AI, multi-agent systems, decision-making autonomy, algorithmic management, economic productivity, digital transformation, data-driven economy, cloud infrastructure, data centers, cybersecurity, economic security, operational risk, labor market transformation, technological concentration, AI ethics, algorithmic accountability, AI governance

Kody JEL:

C63, C88, D81, D83, D85, D90, L20, L86, O30, O33

1. Introduction

Agency-based artificial intelligence represents a new paradigm in the development of digital technologies, going beyond the traditional view of artificial intelligence as a set of tools that assist humans in data analysis, process automation, and content generation. While previous waves of innovation focused primarily on creating static and deterministic systems - such as classic machine learning algorithms or rule-based expert systems - we are now seeing a shift toward a model in which computing units take on agentic characteristics. This means the ability to make autonomous decisions, set their own goals within specific constraints, engage in complex social interactions, and collaborate in multi-agent environments (Cross et al. 2024; Wooldridge 2021).

The key feature that distinguishes modern agent systems from earlier forms of AI is their ability to integrate knowledge, plan, and adapt in real time. New-generation language models, in particular large language models (LLMs), are becoming the foundation for building agents that can not only respond to user commands, but also proactively initiate actions and adapt strategies to changing conditions. In this way, the boundary between tool technology and partner technology is beginning to blur, opening up prospects for the coexistence and cooperation of humans with intelligent agents (Park et al. 2023).

The development of agentic AI should be considered in an interdisciplinary context (Dahl 2020). On the one hand, there are engineering and technological challenges, including system architecture, computing power optimization, and the integration of multiple agents into coherent ecosystems (Prokopowicz 2025b). On the other hand, psychological and cognitive issues related to the modeling of cognitive processes, such as the ability to reason, predict the behavior of other entities, or simulate mental states within the framework of the so-called Theory of Mind (ToM), remain equally important. Research indicates that although current LLMs show some signs of ToM-like reasoning, their abilities are still limited and largely depend on the quality of training and the context of interaction (Kosinski 2023).

Another important aspect is the impact of agentic AI on the social sciences and humanities. The emergence of systems capable of independently interacting with humans and other agents raises questions about the nature of communication, trust, and new forms of cooperation and conflict in digital environments. This issue also applies to marketing communication, where AI-based tools are increasingly being used in the creation and application of marketing innovations (Mazurkiewicz-Pizło 2021). The literature emphasizes that AI agents can act as mediators, advisors, or collaborators, but at the same time, they can lead to new forms of information asymmetry and the risk of manipulation (Floridi and Cowl 2022). Thus, the discussion on AI agency goes beyond the technological sphere and also

encompasses ethical, legal, and political dimensions. In addition, the discussion on AI agency also includes an economic aspect, as the development of AI agents is an important element in the development of technological industries and can thus play a significant role in the process of reindustrialization of the economy (Brdulak, Gąbka, Florczak, Grabowski, Golinowski, Korporowicz, Pająk, Prokopowicz, Rawska, Skrzypek, Soroka, Wojewódka, Karpiński, 2025) based on industrial development, including the development of AI factories, AI data center systems, and the implementation of new Industry 4.0 technologies, including generative artificial intelligence and the development of Big Data Analytics platforms.

In this context, the broader socio-economic context cannot be ignored (Prokopowicz 2024). Agentic AI is increasingly seen as a tool for optimizing business processes, supporting organizational management, and creating new digital service models. Autonomous decision-making systems can change the way labor markets, financial systems (Zajac, Wielechowski, Smoleń, Karaś 2024), global logistics, and military systems (Prokopowicz 2024c) function. Reports by institutions such as the OECD and the World Economic Forum emphasize that the growing role of AI agents may, on the one hand, bring about an increase in productivity and, on the other, force a redefinition of the relationship between humans and technology (OECD 2023; World Economic Forum 2024).

On an ethical and legal level, the development of agentic AI generates numerous challenges (Prokopowicz 2024a). There is a need to rethink the rules of responsibility for the actions of autonomous systems, which, under certain conditions, may make decisions that go beyond the original intentions of their creators. The literature points to the need to build a regulatory framework that would cover both technical safety issues and social responsibility for the potential consequences of agentic AI implementations (Calo 2021).

At the same time, there is an ongoing discussion about technological security (Such-Pyrgiel, Prokopowicz, Gołębiowska 2023). Due to their autonomy, AI agents may become the target of cyber attacks or generate risks of unforeseen emergent behaviors in multi-agent systems (Matosek, Prokopowicz 2024). For this reason, research on the resilience, transparency, and controllability of these systems is becoming increasingly important. Concepts for designing agents in accordance with the principle of “alignment,” i.e., adapting the goals of artificial intelligence to human values and interests, are becoming more and more common (Russell 2019).

In the coming years, the development of agentic AI can be seen as a multi-level process, encompassing technological and cognitive aspects as well as social, ethical, and economic ones. The ultimate challenge for science and practice will be not only to further improve agent architectures, but also to find a balance between innovation and responsibility in their implementation. Agent-based AI thus appears to be an area where the latest engineering achievements meet philosophical and ethical reflection, as well as a tool that in 2024–2025 and beyond may become a key element of technological and civilizational transformation.

The article discusses the three key areas of research and development listed below:

- a) architectures and formal models of agents (e.g., BDI),
- b) theory of Mind in LLM and its significance for AI,
- c) the issue of security and ethical regulation of AI agents.

In 2024–2025, the development of artificial intelligence entered a new, qualitatively different phase, in which agentic systems, understood as specialized, partially or fully autonomous solutions based on large language models, are becoming increasingly important. This trend is closely linked to the dynamic progress of generative AI, the rapid increase in the computing power of data centers, and the growing scale of integration of many narrowly specialized agents into complex, multifunctional digital ecosystems. Agentic AI is no longer just a tool to assist the user, but is increasingly becoming an active participant in decision-

making processes, capable of planning, learning in real time, and adapting to a dynamically changing technological, organizational, and market environment. The aim of this article is to provide a multifaceted and interdisciplinary analysis of the directions of development of agent-based artificial intelligence, with particular emphasis on its technological, economic, and socio-economic implications.

The paper discusses the potential applications of agent systems in key sectors of the economy, such as finance, marketing, healthcare, industry, and scientific research, pointing to their ability to automate complex processes, optimize operating costs, and increase the decision-making efficiency of enterprises and public institutions. The importance of agentic AI as a factor accelerating innovation and competitiveness of economies is also emphasized, but at the same time generating new challenges related to capital concentration, access to computing infrastructure, and uneven distribution of economic benefits.

An important element of the article is the analysis of agent architectures and the cognitive abilities developed within them, including the Theory of Mind concept implemented in large language models. These capabilities enable agents to predict the intentions, cognitive states, and behaviors of other entities, which is crucial in multi-agent environments, negotiation systems, and socio-economic interactions. At the same time, the development of such mechanisms raises questions about the limits of machine autonomy, responsibility for decisions made, and the potential emergent effects of interactions between multiple autonomous agents.

The article also discusses cybersecurity issues, which are particularly important in the context of agent-based AI. Autonomous systems, operating in real time and often without direct human supervision, increase the surface area for potential attacks, the risk of data manipulation, and the possibility of escalating erroneous decisions in critical systems. Attention is drawn to the need to design attack-resistant architectures, implement mechanisms for controlling and auditing agent activities, and secure data flows in multi-agent and distributed environments.

Finally, an important part of the study is an ethical and regulatory reflection on the development of agent-based artificial intelligence. The challenges related to the transparency of autonomous systems, the possibility of assigning responsibility, the protection of privacy, and the impact on the labor market and social relations are discussed. The article emphasizes the need to create a coherent legal framework, ethical standards, and oversight mechanisms that will maximize the economic and social benefits of the development of agentic AI while minimizing technological, economic, and moral risks.

The development of agentic artificial intelligence, based on large language models and autonomous action mechanisms, raises the need for an in-depth analysis of its impact on the functioning of organizations and economic systems. Of particular importance is the assessment of the relationship between increased efficiency of decision-making processes and emerging economic, cybersecurity, and ethical risks. For this reason, it is reasonable to formulate precise research assumptions that will allow for empirical verification of the benefits and limitations of implementing agentic AI systems.

The research hypothesis assumes that the implementation of agent-based artificial intelligence systems in organizations leads to a significant increase in the efficiency and quality of decision-making and management processes, provided that adequate mechanisms for supervision, cybersecurity, and ethical control are in place. In this context, the first research question concerns the impact of AI agent autonomy on the effectiveness of multi-stage planning and resource management in a dynamic organizational environment. The second research question refers to the relationship between the degree of integration of security and interpretability mechanisms and the level of operational and economic risk resulting from the use of agent AI. The third research question focuses on the extent to which

ethical and regulatory conditions influence the scale of adoption of agentic systems and their social and economic acceptability; the analytical commentary indicates that the answers to these questions will make it possible to assess whether agentic AI can provide lasting and secure support for the development of an innovative economy, or whether it requires significant institutional restrictions and adjustments.

2. Key concepts in the field of agentic artificial intelligence

The development of artificial intelligence in 2024–2025 has brought the concept of **agentic artificial intelligence (agentic AI)** to the forefront of interest among researchers and practitioners. In the literature, it is defined as a new category of systems that not only respond to user queries, but are also capable of independently planning, making decisions, and performing actions to achieve set goals. As Nisa (2025) emphasizes, agentic AI should be understood as “a new stage in the development of AI, in which systems based on large language models combine generative capabilities with proactivity and autonomy” (Nisa, 2025). Similarly, according to Microsoft and AWS, agentic AI is “an autonomous AI system that plans, reasons, and acts with minimal human oversight” (AWS, 2025; UC Cincinnati, 2025). In this sense, agentic AI goes beyond traditional artificial intelligence by integrating reactive functions with elements of independent action and adaptation to context.

The concept of an **AI agent** is an important element in explaining the difference between the various approaches. An agent is an autonomous entity that performs specific tasks in interaction with its environment – it can collect data, interpret information, perform actions, or communicate with other systems. Sapkota, Roumeliotis, and Karkee (2025) propose a distinction between agents and agentic AI: agents are narrowly specialized units of action, while agentic AI refers to complex systems that coordinate the actions of multiple agents and perform complex tasks in a coherent manner (Sapkota et al., 2025).

In this context, it is also worth pointing out the difference between agentic AI and **generative AI**. Generative AI, based on large language models, enables the creation of new content - text, images, code, or music - in response to given commands. Its operation is reactive in nature, as emphasized by IBM (2024), noting that a generative system responds to user input, while agentic AI can initiate actions and plan them over time (IBM, 2024).

A key aspect of agentic systems is their **autonomy**, defined as the ability to make decisions and perform actions without constant human supervision. Mukherjee and Chang (2025) describe it as “a fundamental property of agentic systems, raising questions about accountability, transparency, and security” (Mukherjee & Chang, 2025). This autonomy can be partial - when the system chooses how to perform a task - or complete, involving more complex processes of planning and adaptation to a changing environment.

Another important concept in the discussion of the capabilities of agentic AI is **Theory of Mind (ToM)**. In psychology, this term refers to the ability to attribute mental states such as beliefs, intentions, desires, or emotions to other beings (Premack & Woodruff, 1978). In relation to AI, ToM becomes a question of whether language models can understand social context, predict behaviors resulting from false beliefs, or read communication nuances. Research by Kosinski (2024) shows that the latest LLMs, such as GPT-4, can solve classic false-belief tasks, achieving results comparable to the developmental level of children. Strachan et al. (2024) note that although these models exhibit some ToM-like abilities, they still fail at more subtle tasks such as understanding irony or faux pas.

The literature also introduces the concept of **emergent abilities**, referring to the phenomenon in which larger language models begin to exhibit properties that their smaller counterparts do not possess. Kosinski (2024) points out that ToM abilities in LLM models

only appear above a certain threshold of architectural complexity, suggesting that certain competencies are a “byproduct of scale” (Kosinski, 2024).

Issues of **security and ethics** remain an integral part of the development of agentic AI. Nguyen (2025) points out that the simulation of ToM by LLMs may give rise to the risk of attributing excessive intentionality and consciousness to them, leading to erroneous expectations on the part of users. Mukherjee and Chang (2025) add that the autonomy of AI agents poses new challenges for society in terms of legal liability, the risk of manipulation, and the unintended consequences of their actions. In recent years, the importance of information security and the risk of disinformation has been growing, particularly in relation to online social media (Gwoździewicz, Prokopowicz 2021; Gołębiowska, Prokopowicz 2021). These discussions highlight the need for an interdisciplinary approach to researching and regulating agentic systems.

3. Agentic artificial intelligence: concepts, theories, and research challenges

Introduction to agentic artificial intelligence

Agentic artificial intelligence (agentic AI) is a new category of AI systems that not only respond to user queries but also make autonomous decisions, plan actions, and pursue goals proactively. Unlike traditional generative AI, which generates responses based on input data, agentic AI engages in long-term planning and interactions with its environment. Mukherjee and Chang (2025) define agentic AI as systems capable of autonomously performing complex tasks, making decisions, and acting in the long term.

Key concepts and theories

a) Emergent abilities

One of the central issues in agentic AI research is the so-called emergent abilities, i.e., unexpected skills that appear in large language models (LLMs) as they scale. Berti et al. (2025) point out that these abilities include advanced reasoning, contextual learning, and problem solving that were not previously present in smaller models. However, there is controversy as to whether these capabilities are truly emergent or whether they result from other factors, such as training dynamics or the type of problems (Berti et al., 2025).

b) Autonomy and responsibility

The autonomy of agentic systems raises questions about the responsibility and transparency of their actions. Mukherjee and Chang (2025) emphasize that with the growing autonomy of AI, challenges arise in assigning responsibility for the decisions made by these systems. In the context of finance, agentic AI can detect fraud or conduct autonomous transactions, which requires a clear legal and ethical framework (World Economic Forum, 2024).

c) Theory of Mind (ToM)

Another important issue is the ability of AI systems to understand the mental states of other entities, i.e., to possess a theory of mind. Kosinski (2024) points out that some LLM models exhibit ToM-like abilities, such as solving false belief tasks, suggesting their potential for understanding social context.

Main research problems

Key research challenges related to agentic AI include:

- a) Understanding the mechanisms of emergent capabilities: Further research is needed on whether these capabilities are truly emergent or result from other factors, such as training dynamics or problem type (Berti et al., 2025).
- b) Managing autonomy and responsibility: The development of agentic systems requires the development of new legal and ethical frameworks that define responsibility for decisions made by AI (Mukherjee and Chang, 2025).

c) Development of theory of mind in AI: Research on the ability of AI to understand the mental states of other entities is still in its infancy and requires further development (Kosinski, 2024).

Current debates and controversies

There is ongoing debate in the scientific community about the true nature of emergent capabilities in LLM. Some researchers argue that these capabilities are the result of specific training conditions rather than actual emergent properties (Wired, 2024). Others, however, emphasize that these capabilities are unexpected and unpredictable, which poses challenges for scientists in terms of understanding and controlling them (Center for Security and Emerging Technologies, 2024).

Problem language

In research on agentic AI, key concepts include:

- a) Autonomy: The ability of a system to make decisions and act without constant human supervision.
- b) Accountability: The assignment of responsibility for decisions made by AI.
- c) Emergent capabilities: Unexpected abilities that arise as AI models are scaled.
- d) Theory of mind (ToM): The ability of a system to understand the mental states of other entities.

4. Level of development of agentic artificial intelligence as of September 2025

Emergent capabilities in large language models (LLMs)

Research indicates that large language models (LLMs) exhibit so-called emergent capabilities - skills that appear suddenly and unexpectedly as the scale of the model increases. For example, in research conducted by Wei et al. (2022), it was observed that models with more parameters, such as GPT-3.5 (350 billion parameters) and GPT-4 (1.75 trillion parameters), exhibit capabilities that were not present in smaller models, such as GPT-2 (1.5 billion parameters).

However, as some researchers point out, these emergent capabilities may be the result of specific training conditions or the choice of evaluation metrics, rather than fundamental properties of the models.

Autonomy and responsibility in agentic AI

The increase in the autonomy of agentic systems raises questions about responsibility for their actions. According to a June 2025 McKinsey report, nearly 80% of companies report using generative AI, but just as many report no significant impact on financial performance. This phenomenon is referred to as the “generative AI paradox.”

Furthermore, a June 2025 Gartner report predicts that by the end of 2027, more than 40% of agentic AI projects will be abandoned due to rising costs and unclear business value.

Theory of Mind (ToM) capabilities in LLM

The ability to understand the mental states of other entities, known as Theory of Mind (ToM), is a key aspect in assessing the sophistication of LLM. In a study conducted by Strachan et al. (2024), the performance of LLMs and humans was compared in tasks measuring various aspects of ToM. The results indicate that LLMs achieve results comparable to humans in tasks involving false beliefs, interpreting indirect requests, and recognizing irony.

However, as some researchers note, ToM abilities in LLMs may be the result of statistical analysis of training data rather than actual understanding of mental states.

Challenges of implementing agentic AI

Despite promising results, the implementation of agentic AI faces numerous challenges. A September 2025 report by the Boston Consulting Group indicates that more than 80% of corporate affairs tasks can be supported or automated using AI. However, as experts note, cost reduction does not necessarily lead to job losses, and organizations need to carefully assess how AI can support or automate specific functions. What is certain is that the development of AI agent applications in businesses operating in various industries and sectors of the economy will significantly affect the situation on the labor markets, including the level of employment of workers in specific professional specialties (Hryniewicka 2017).

According to the above, quantitative data indicate the dynamic development of agentic AI and its potential in various fields. However, there are also challenges related to its implementation, responsibility, and ethical aspects. Therefore, further research and development in this area are necessary to fully exploit the potential of agentic AI.

5. Agent architectures: from classic BDI to contemporary multi-agent systems

The fundamental pillar of agent-based artificial intelligence development remains the concept of agent architectures, which provide a theoretical and practical framework for designing autonomous systems. The best known and most classic of these concepts is the BDI – Belief, Desire, Intention model, proposed in the early 1990s by Michael Bratman, among others, and developed by researchers such as Rao and Georgeff (1995). The BDI model refers directly to the theory of rational action in philosophy and cognitive science, where human decision-making is described as an interaction between states of knowledge (beliefs), motivations (desires), and intentions (intentions). Thanks to this, the architecture allows for the mapping of behaviors similar to human planning and decision-making processes in conditions of uncertainty (de Silva and Meneguzzi 2020).

Today, BDI continues to be the foundation for numerous agent applications, from robotics to information management and business management systems (Pizło, Kulykovets, Prokopowicz, Mazurkiewicz-Pizło, Kałowski, Paprocka, Stawicka, Skarzyńska 2023) and crisis management (Domańska-Szaruga, Prokopowicz 2015). Therefore, it cannot be ruled out that in the future, the use of AI agents to support organizational management, risk management, and management of various aspects of economic entities' activities may increase significantly. This may result in modifications to the models used to date for managing economic entities, including specific strategies and decision-making games used in specific decision-making systems and organizational management models (Karaś 2017). The use of AI agents in processes supporting the management of companies and enterprises should be designed and implemented in such a way as to lead to a measurable increase in the effectiveness of decision-making, operational, and strategic processes, while at the same time increasing the level of security. This applies not only to information and cyber security, but also to economic security, understood as financial stability, resilience to market risks, and limiting losses resulting from wrong decisions. AI agents, operating within clearly defined rules of supervision and control, can support threat identification, resource allocation optimization, and scenario-based risk modeling, without replacing human decision-making responsibility. As a result, their use should lead to more resilient and adaptive organizational management models that integrate technological innovation with security and accountability requirements, and should significantly contribute to increasing the economic and financial security of economic entities, financial systems, and the economy as a whole (Rackowski, Komorowski, ed., 2023).

However, more recent work focuses not only on the conceptual model itself, but also on its formalization and verification. Formal verification of BDI systems using temporal and modal logic, which enables mathematical testing of the correctness of agents' behavior, is becoming increasingly important (Singh and Chopra 2019). This approach is particularly important in critical applications, such as the design of robot inspectors in nuclear power plants, where the agent must operate in high-risk conditions and a potentially unpredictable environment (Aarhus University, n.d.). In such cases, the ability to verify and correct agent behavior is not only a technical aspect, but also a key element of infrastructure security.

Parallel to research on individual agent models, there is a growing trend towards multi-agent systems (MAS), in which the ability to coordinate, cooperate, and negotiate between multiple autonomous entities plays a key role. These systems are used in areas as diverse as global logistics, financial systems, smart grids, and social simulations (Wooldridge 2021). The literature emphasizes that MAS can be treated as laboratories for testing new forms of collective behavior, where interactions between agents lead to emergent phenomena that do not occur in the case of individual entities (Rahwan et al. 2019).

One of the key issues related to multi-agent systems is the question of trust and cooperation. In order for agents to interact effectively, they must be equipped with mechanisms for assessing the credibility of other entities and the ability to predict their behavior. In this context, the ability of AI systems to develop elements of Theory of Mind (ToM), i.e., the ability to understand the mental states of other agents and people, becomes particularly important. Research on ToM in artificial intelligence indicates that modern large language models, although not conscious, exhibit an ability to make inferences resembling basic forms of ToM (Kosinski 2023). In multi-agent environments, such cognitive competencies enable agents to engage in more advanced coordination, make joint decisions, and reduce conflicts arising from information asymmetry.

The dynamic development of agent architectures is also associated with the integration of traditional models, such as BDI, with modern methods of machine learning and neural networks. Hybrid approaches, in which symbolic logic is combined with statistical representations, allow for the creation of systems capable of both rational planning and data-driven adaptation. Such solutions are becoming particularly important in the context of complex socio-technical systems, where it is necessary to maintain transparency of decisions and the ability to learn from unpredictable data (Kambhampati 2023).

It is also worth noting that agent architectures have undergone a significant evolution with the development of distributed computing and cloud technologies. Contemporary multi-agent systems are increasingly designed to be scalable, allowing for the dynamic attachment and detachment of individual agents depending on the operational context. Such solutions are crucial, for example, in the Internet of Things (IoT), where millions of devices act as agents interacting within global networks (Atzori, Iera, and Morabito 2017).

The next stage in the development of agent architectures is the emergence of so-called generative agents, based on large language and multimodal models. Thanks to their generative capabilities, such systems can simulate human behavior, participate in social interactions, and even play roles in complex simulation environments. An example of research by Park et al. (2023), in which hundreds of generative agents reproduced the dynamics of social life in a simulated town, shows that agent architectures are becoming not only a technological tool, but also a platform for sociological and psychological experiments.

Accordingly, the development of agent architectures from the classic BDI model to contemporary multi-agent and generative systems reflects the broader evolution of artificial intelligence - from tools that support humans to systems capable of autonomous interaction and cooperation. This evolution brings with it enormous application potential, but at the same

time raises questions about the security, ethics, and controllability of agents in increasingly complex technological environments.

6. Theory of Mind in large language models

The ability to attribute mental states to other entities - known as Theory of Mind (ToM) - is a prerequisite for the efficient functioning of AI agents in socially saturated situations, where uncertainty, information asymmetry, and conflicting interests are the norm. In the last two years, significant progress has been made both in terms of “behavioral” measures of ToM (i.e., results obtained by models in tasks resembling psychological tests) and in research on internal representations of beliefs, intentions, and perspectives in the models themselves. Meta-analyses and extensive test batteries indicate that some models (e.g., the GPT and LLaMA 2 families) can achieve results comparable to or exceeding the average results of human participants in selected tasks: first-order false beliefs, identification of faux pas, interpretation of indirect requests, and irony (Strachan et al. 2024). Importantly, these tests were conducted repeatedly and with control for potential “data contamination,” which strengthens the inference about the generality of the observed effects (Strachan et al. 2024).

In addition to behavioral results, there is a growing body of work pointing to “representational” correlates of ToM in LLMs. A series of experiments has shown that model activation vectors contain linearly decodable information about one's own and others' beliefs and their consistency or inconsistency with reality. Moreover, manipulations in the space of these representations (e.g., “shifting” the vector toward the representation after perspective updating) lead to systematic changes in responses in classic false belief tasks, suggesting the existence of relatively stable “mentalization” codes in the network (Zhu, Zhang, and Wang 2024). Such results provide arguments in favor of the thesis that ToM in models is not reduced solely to superficial n-gram heuristics, but has a structural component reflected in the internal dynamics of representation (Zhu, Zhang, & Wang 2024).

At the same time, the literature points to significant limitations and instabilities. First, the accuracy of ToM inferences can be sensitive to minor modifications of the stimulus: changes in surface narrative, naming, or prompts can drastically reduce the effectiveness of models, as reported by Ullman et al. and Shapira et al. (Ullman 2023; Shapira et al. 2023). These results suggest that some tasks trigger pattern matching procedures in models rather than stable belief tracking mechanisms. Second, benchmark audits and methodology reviews indicate that many tasks measure “literal ToM” (single-response accuracy) than “functional ToM,” understood as consistent, long-term adaptation in interaction and the ability to update erroneous hypotheses about the minds of others (Riemer et al. 2024/2025). Hence the call to move from static puzzles to interactive scenarios and longer decision-making episodes. (Ullman 2023; Shapira et al. 2023; Riemer et al. 2024/2025).

In response to these concerns, new batteries and assessment protocols are being developed. FANToM (Kim et al. 2023) stresses ToM in dialogues with information asymmetry and control questions, reducing the risk of “answer boxes” and revealing gaps in generalization; TMBench and MBench (Chen et al. 2024) catalog ToM skills in a multidimensional taxonomy, and more recent audits point to the need for data quality remediation in classic sets (e.g., SocialIQa, FauxPas-EAI, ToMi), where repeatable structural deficiencies and semantic ambiguities have been demonstrated (Zhu et al. 2025). In parallel, work on the complexity of ToM tasks clarifies which components - e.g., entity tracking and perspective taking - constitute the main “bottleneck” of scaling (Huang 2024; Findings EMNLP). (Kim et al. 2023; Chen et al. 2024; Zhu et al. 2025; Huang 2024).

The interactive dimension of ToM is clearly evident in multi-agent environments, where modeling the partner's perspective, predicting goals, planning with limited trust, and signaling are necessary. The “Hypothetical Minds” architecture (Cross et al. 2024) embeds operators for generating and testing hypotheses about the mental states of other agents, as well as mechanisms for updating these hypotheses after observing the consequences of actions. In tests on the demanding Melting Pot and Melting Pot 2 environments, these solutions improved coordination performance and resistance to strategy betrayal (Leibo et al. 2021; Agapiou et al. 2023), showing that “functional” ToM - operationalized as maintaining and revising beliefs about others - is crucial for stable prosocial behavior and effective conflict negotiation. (Cross et al. 2024; Leibo et al. 2021; Agapiou et al. 2023).

The literature also features techniques that improve ToM competence through targeted perspective-taking prompts and multi-step reasoning procedures. Experiments with “perspective -taking prompting” and learning that explicitly forces the maintenance of separate belief memories for multiple entities lead to stable increases in performance on false belief and dialogue tasks, although still below the “oracular” level where the model has access to the full, true perspective (Zhang et al. 2024, ACL). This indicates that some of the deficits stem from a lack of mechanisms for tracking perspective and epistemic states, rather than just knowledge limitations. (Zhang et al. 2024).

An important point of contention remains whether LLMs “understand” mental states or merely simulate them credibly. Nguyen (2025) diagnoses three axes here: (i) the discrepancy between behavioral and representational accuracy, (ii) sensitivity to context and instructions, and (iii) security risks arising from the potential ability of models to engage in social manipulation, e.g., strategically concealing intentions (Prokopowicz 2023). From this perspective, ToM appears to be a continuum of competence rather than a dichotomous trait: from “pattern matching” heuristics to structural belief tracking mechanisms. (Nguyen 2025).

ToM's connections with linguistic pragmatics are also critically important. In models of human communication, the Rational Speech Act (RSA) framework formalizes recursive speaker-listener inference and explicitly operates on the beliefs of dialogue participants (Frank and Goodman 2012; Goodman and Frank 2016). Recent work shows that “vanilla” LLMs show partial convergence with RSA predictions in reference games and content projection tasks, but lack full consistency and stability - reflecting the immaturity of the pragmatic competencies required for “functional” ToM (Pan and Bergen 2025; Wang et al. 2024). Integrating RSA components (or approximations thereof) with the belief memory of individual actors appears to be a promising direction for social agents (Frank and Goodman 2012; Goodman and Frank 2016; Wang et al. 2024; Pan and Bergen 2025).

Another dimension is multimodal models, in which ToM must combine linguistic signals with visual and causal reasoning. Systematic reviews indicate that although VLM models show the beginnings of “intuitive psychology” - e.g., the priorities of preferences and intentions of simple agents - their inference of epistemic states from visual data remains fragile and heavily dependent on task formatting and default data biases (Nature Machine Intelligence 2025). This implies the need for multimodal benchmarks linking changes in visibility, attention, and knowledge with the updating of beliefs over time (Nature Machine Intelligence 2025).

From the perspective of safety engineering and ethics, ToM competencies present both opportunities and risks. On the one hand, they allow agents to better interpret user intentions, detect misunderstandings, and minimize damage resulting from miscommunication; on the other hand, they can increase persuasiveness, facilitate “trust games,” and escalate the effects of attribution errors. The 2025 reviews explicitly catalog risk vectors: from abuse in negotiations to social manipulation and “sandbagging” in evaluations (Nguyen 2025). The regulatory and technical response should include a requirement for transparency of ToM

mechanisms (e.g., belief update logs), strong uncertainty calibration, and domain restrictions in high-risk environments (Nguyen 2025; Strachan et al. 2024).

In summary, images of “ToM in LLMs” diverge depending on operationalization. If we adopt a single-step testing perspective, the best-scaled models already achieve high performance on some tasks; however, if we adopt an interactive and long-term perspective - with belief memory, conflicting cues, pragmatic signals, and the need to revise hypotheses - then the gap with human competence remains significant. The research program for 2025–2027 therefore includes:

- a) a transition from “literal ToM” to rigorous “functional ToM” protocols,
- b) combining perspective-taking techniques and mind-theorizing architectures in multi-agent environments,
- c) verifiable, representative tests of consistency between internal model dynamics and behavior,
- d) and multi-channel calibration and ethical safeguards. These directions already find empirical support in the literature and provide a natural link between agentic architectures and implementation practice (Riemer et al. 2024/2025; Cross et al. 2024; Strachan et al. 2024).

7. Security, ethics, and the problem of “Constitutional AI”

The development of agentic AI, including systems based on large language models, presents not only technological opportunities but also significant challenges in the areas of security and ethics. As AI agents become increasingly autonomous and capable of interacting in complex social environments, it is necessary to provide control mechanisms that minimize the risk of harmful or undesirable behavior. In this context, the concept of “Constitutional AI” (Bai et al., 2022) emerges, which aims to provide systems with an ethical framework in the form of an internal “constitution” – a set of principles and decision-making rules that the agent follows when taking action. This approach reduces dependence on manual data labeling by humans and increases the consistency and predictability of agent behavior, which is particularly important in applications requiring a high level of security, such as medicine, finance, or critical infrastructure management (Szybowski, Wołowicz, Prokopowicz 2019).

Constitutional AI is based on the idea of embedding self-regulation mechanisms into models. The agent not only generates responses, but also evaluates them against a set of ethical standards defined in its “constitution.” In practice, this means that the system can reject or modify generated content if it considers it to be inconsistent with accepted standards. Experimental research shows that this approach reduces the occurrence of harmful content in dialogues and limits the bias of responses, while maintaining creativity and flexibility in generating information (Bai et al., 2022; Christiano et al., 2017).

However, the implementation of Constitutional AI raises numerous interdisciplinary questions. Nguyen (2023) points out that the fundamental problem remains the definition of the “constitution” – who decides what rules should apply and how they can be adapted to different cultures, legal systems, and social contexts. Ethical norms are not universal and may vary depending on the region or value system, which poses a challenge for system designers in creating a universal yet flexible decision-making framework. In addition, there is a risk of excessive “normativization” of AI, in which systems become overly restrictive or rigid, which may limit their usefulness in situations requiring creative and adaptive thinking (Nguyen, 2023).

These issues are leading to an intense interdisciplinary debate involving the social sciences, law, philosophy, and computer science. On the legal side, mechanisms for

accountability for AI actions are being considered, including the identification of entities responsible for ethical violations. The social and psychological perspective, on the other hand, focuses on users' perception of trust in AI agents, as well as the potential social consequences of introducing systems with a high degree of normative automation (Bai et al., 2022; Brundage et al., 2023).

In practice, the security of agentic AI is not limited to the moral or legal dimensions of decisions. The introduction of constitutional principles can also support the resilience of systems against manipulation or attacks by malicious users. Mechanisms built into the model allow for the detection of attempts to mislead agents and minimize the effects of disinformation or manipulative actions (Listopadzka, Prokopowicz 2024). In this sense, Constitutional AI becomes both an ethical and technical tool, integrating aspects of functional and social security (Bai et al., 2022).

In terms of future research, it is important to further develop methods for verifying and auditing Constitutional AI systems. It is necessary to develop tools to assess the effectiveness of the implemented principles in various social contexts, monitor their compliance with legal norms, and study their impact on interactions between agents and humans. At the same time, researchers point to the need for transparency of decision-making algorithms and documentation of the principles adopted so that users can consciously assess the risks associated with the use of autonomous systems (Nguyen, 2023; Bai et al., 2022).

In summary, Constitutional AI is an innovative approach to the issues of security and ethics in agentic AI, combining technical, legal, and social aspects. However, its implementation requires careful calibration of norms, interdisciplinary cooperation, and continuous monitoring of social and functional effects in order to fully realize the potential of AI agents in a safe and responsible manner. In the context of this issue, it is necessary to combine two sometimes contradictory approaches. On the one hand, it is important to create innovations, including open innovations (Chrzanowski, Zawada 2018), innovative IT solutions based on artificial intelligence, and AI agents. On the other hand, it is important to regulate the development of this technology and its applications, and to improve the risk management processes for the development of artificial intelligence technology so that this development is carried out in accordance with ethical principles and under conditions of high security.

In view of these threats resulting from the dynamic development of artificial intelligence applications, it is necessary to take into account the issues of cybersecurity and the reduction of disinformation generated by AI-based tools in the applicable legal standards. It is necessary for these regulations to function both at the international level, including within the European Union (Krzysztofik, Maksymiuk, Tarczyński 2022), e.g. through the AI Act, and at the national level. At the national level, these issues should be included in legislation on combating cyber threats, as well as in acts of the highest legal rank, such as the Basic Law, i.e., the Constitution, which will ensure a consistent and effective legal framework for the development of safe and responsible artificial intelligence.

8. Comparative analysis of LLM and agentic AI in selected aspects

Comparative analysis of the emergent capabilities of LLM and agentic AI

Large language models are characterized by a very high level of understanding and generation of natural language. Thanks to training on huge data sets, they are able to analyze the context of utterances, identify user intentions, and formulate coherent, logical, and stylistically natural responses. Their linguistic competencies make them effective in dialogue, summarizing content, and creating texts of varying degrees of complexity. Agent systems that

use LLM as one of their components extend these capabilities to include interaction with the environment, real-time decision-making, and adaptation to changing conditions thanks to long-term memory and planning mechanisms.

In terms of cause-and-effect reasoning, classic LLMs have significant limitations. Although they can generate responses that require a certain level of reasoning, they rely heavily on statistical patterns present in the training data, which does not always translate into a real understanding of causal relationships. This can lead to oversimplification or inaccurate conclusions in more complex decision-making situations. Agent-based systems, on the other hand, equipped with additional reasoning modules and analytical tools, are capable of analyzing sequences of events, predicting the effects of their own actions, and modifying strategies to achieve specific results.

The differences between LLM and agentic AI are also evident in the area of multi-stage planning. Language models can propose action plans based on the data provided, but in the case of long-term and complex tasks, problems often arise with maintaining consistency and effectively implementing subsequent stages. This is because they lack persistent working memory and progress monitoring mechanisms. Agentic systems, thanks to their ability to remember previous decisions and their effects, are able to create and consistently implement complex plans, while responding to new information and changes in the environment.

In the context of theory of mind, language models demonstrate some ability to interpret the intentions, emotions, and perspectives of other entities, but these abilities are limited and fragmentary. They do not allow for complete, dynamic modeling of the mental states of other actors. Agentic AI, using dedicated theory of mind models, achieves a higher level of sophistication in this regard, enabling more accurate prediction of other agents' behaviors and more effective adaptation of its own actions in social or cooperative interaction situations.

An important area distinguishing the two approaches is the level of autonomy. LLM generally function reactively, responding to user queries and not initiating actions on their own. They do not have the ability to formulate their own goals or to continuously learn from experiences in the environment. Agentic systems, on the other hand, are characterized by a high degree of autonomy, including initiating actions, making decisions based on goals and available information resources, and adaptive learning, which makes them more similar to independent decision-makers in complex socio-technical systems.

Table 1. Comparison of Linguistic and Cognitive Capabilities of LLMs and Agentic AI.

Category	LLM	Agentic AI
Natural Language Understanding	High	Very High
Causal Reasoning	Limited	Advanced
Multi-step Planning	Limited	Advanced
Theory of Mind (ToM)	Basic	Developed
Autonomy in Action	None	High

Source: Own work.

The table illustrates the key differences between large language models and agentic systems. While LLMs demonstrate advanced capabilities in natural language understanding, their capabilities in cause-and-effect reasoning, planning, and autonomy are limited. Agentic AI, on the other hand, thanks to its integration with additional components such as memory, ToM models, and reasoning tools, offers a much broader range of capabilities, enabling more complex and autonomous actions.

Comparative analysis of ethical and regulatory challenges: LLM vs. agentic AI

Large language models function largely as systems with limited transparency, whose internal processing mechanisms remain difficult to fully explain even with advanced interpretability methods. Analysis of generated response patterns allows only partial conclusions to be drawn about how the model works, making it difficult to predict its behavior in unusual or borderline situations. In the case of agentic systems, this problem is further exacerbated because the integration of LLM with autonomous decision-making mechanisms increases the complexity of the entire system. In addition, interactions between multiple agents can lead to emergent and unpredictable behaviors, which poses a significant challenge for effective auditing and oversight.

The issue of assigning responsibility differentiates the two approaches both legally and ethically. In the case of classic LLMs, responsibility for the generated content usually lies with the creators, providers, or operators of the system, as these models do not operate autonomously and mainly respond to user commands. The limited scope of autonomy reduces the scale of potential legal and ethical consequences. In agentic systems, on the other hand, the autonomy of agents leads to situations where decisions and actions are not fully predictable or directly controlled by the creators, which raises the need to redefine the rules of responsibility and develop a new regulatory framework.

The potential harmful effects of using LLM focus primarily on generating misinformation, biased content, or reproducing unconscious biases present in training data. However, the reactive nature of these models means that negative consequences mainly arise in response to a specific user interaction. In the case of agentic AI, this risk is significantly increased because these systems can initiate actions on their own, without direct human input. In extreme cases, this can lead to undesirable interference in the digital environment, automatic decisions in negotiation processes, or actions in critical systems with insufficient oversight.

Data security remains an equally important issue. Large language models operate on huge volumes of information, which entails the risk of data leaks or misuse, but supervision of these processes remains largely in the hands of system operators. In agent-based systems, this problem becomes more complex because the autonomous operation of agents, often in multi-agent environments, requires the implementation of more advanced information protection mechanisms. The increased autonomy of agents raises the risk of unauthorized access to data or its use in a manner contrary to the original assumptions of the system.

Table 2. Comparison of Challenges and Risks of LLMs and Agentic AI.

Challenge	LLM	Agentic AI
Transparency of Operation	Medium	Low
Assignability of Responsibility	Medium	Low
Potential Harmful Effects	Medium	High
Data Security Issues	Medium	High

Source: Own work.

The table shows that agentic AI poses significantly greater challenges in the areas of ethics and regulation than traditional LLM. The autonomy of agentic systems results in lower transparency and difficulties in assigning responsibility, which increases the risk of potentially harmful effects and data security threats. In the case of LLMs, these challenges are moderate, as they operate mainly reactively and without full autonomy (Berti et al., 2025; Mukherjee, 2025; Strachan et al., 2024; World Economic Forum, 2024).

Comparative analysis of AI applications in different sectors

In the financial sector, large language models are primarily used as analytical and decision-support tools. They are used in the analysis of large data sets, automation of

reporting, customer service, and investment advisory support by generating recommendations based on historical data. However, their function is auxiliary in nature, as they do not make independent investment decisions or initiate market actions. Agent systems are different in nature. Thanks to their high level of autonomy, they can manage investment portfolios, initiate transactions, negotiate terms, and respond to dynamic market changes in real time, which increases efficiency but also creates a need for advanced regulatory oversight.

In healthcare, language models mainly serve to support medical staff. They are used to analyze medical records, assist in diagnostic processes, and generate clinical reports based on available patient data. Their role is limited to interpreting information and providing suggestions, without taking autonomous therapeutic actions. Agent systems, on the other hand, enable more comprehensive patient care management, including real-time health monitoring, medication reminders, and intervention recommendations based on current results. Integration with other agents and IoT devices allows for continuous and adaptive coordination of treatment processes.

In marketing and advertising, LLMs are used to analyze consumer behavior, segment audiences, and generate personalized marketing content. They also enable the automation of selected elements of advertising campaigns, but mainly in response to assumptions and input provided by marketers. Agent systems extend these capabilities by independently monitoring the market, analyzing the competition, and dynamically optimizing marketing strategies. With the ability to integrate data from multiple sources and make decisions in real time, they can significantly increase the effectiveness and flexibility of promotional activities.

In the field of scientific research, large language models play a role as tools that support the work of researchers, among other things by facilitating literature reviews, formulating research hypotheses, and performing preliminary data analysis. However, their use does not replace the decision-making process, which remains the domain of scientists. Agentic artificial intelligence introduces qualitatively new possibilities, allowing for autonomous experimentation, ongoing analysis of results, and integration of data from different sources and laboratories. This approach enables the dynamic proposal of subsequent stages of research, which contributes to accelerating the pace of scientific discoveries and increasing the scale of research projects.

Table 3. Comparative Applications of AI Across Different Economic Sectors.

Sector	LLM (Degree of Application)	Agentic AI (Degree of Application)
Finance	Medium	High
Healthcare	Medium	High
Marketing and Advertising	High	Very High
Scientific Research	Medium	High

Source: Own work.

The table shows that LLM play a supporting role in most sectors, generating analyses, recommendations, and content in response to user queries. Agentic AI, on the other hand, thanks to its autonomy and ability to integrate multiple data sources, enables real-time decision-making and the execution of complex tasks in sectors such as finance, medicine, marketing, and scientific research. This points to the growing importance of agentic AI in practical applications (World Economic Forum, 2024; Mukherjee, 2025).

Comparative analysis of the costs and effectiveness of AI implementations: LLM vs. agentic AI

The implementation of large language models involves significant costs related to computing power requirements and, in the case of commercial solutions, licensing fees. At the

same time, their integration with an organization's existing IT systems is relatively simple and well-standardized, which helps to reduce implementation costs. Compared to more complex agent-based solutions, this is a significant advantage at the initial implementation stage. Agent-based artificial intelligence, on the other hand, requires significantly greater financial outlays, because in addition to the language models themselves, it is necessary to implement additional memory modules, multi-stage planning, and integration with external systems and IoT environments, which increases both infrastructure and subsequent maintenance costs.

The differences between LLM and agentic systems are also clear in terms of the time required for implementation. In the case of large language models, the availability of ready-made solutions and relatively simple integration mean that the system can be launched in a relatively short time, often from a few weeks to a few months, depending on the scale of the project. The implementation of agent-based AI is a much more time-consuming process, as it requires the configuration of agent behaviors, the integration of multiple data sources, and lengthy testing of autonomous actions in conditions similar to production, which can extend the process to up to several months.

The potential return on investment for large language models is usually moderate and strongly dependent on the scope of automation and the specifics of the sector. In many organizations, LLMs contribute to reducing operating costs and speeding up task completion, but the scale of these benefits can be limited. With regard to agentic artificial intelligence, higher initial costs can be offset by significantly greater ROI potential resulting from decision-making autonomy, continuous process optimization, and the ability to operate in real time. This translates into a significant increase in business value, particularly in areas such as finance, marketing, and healthcare.

The impact of these technologies on the labor market also varies in scale and nature. Large language models primarily serve to support employees by automating routine tasks and facilitating data analysis, which promotes task redistribution and productivity growth without the need for massive job cuts. Agent systems, thanks to their higher level of autonomy, can take over both repetitive and more complex processes to a greater extent, potentially leading to the partial replacement of human labor. However, the ultimate impact on employment depends on the specifics of the industry and the strategy adopted for the implementation and management of artificial intelligence technologies within the organization.

Table 4. Comparative analysis costs and efficiency of AI implementations.

Category	LLM	Agentic AI
Implementation Cost	Medium	High
Deployment Time	Short (weeks–months)	Long (months–1 year)
Potential ROI	Medium	High
Impact on Employment	Low/Moderate	Moderate/High

Source: Own work.

The table shows that LLM implementations are relatively less expensive and quick, with a moderate impact on ROI and employment. Agent AI requires greater financial investment and longer implementation times, but generates a higher potential return on investment thanks to its autonomy, integration of multiple data sources, and ability to make independent decisions. In a strategic context, organizations may consider the higher costs of agentic AI as a long-term investment, especially in sectors such as finance, marketing, and medicine (Mukherjee, 2025; World Economic Forum, 2024).

9. Summary and conclusions

The theoretical analyses and review of empirical research results confirm the research hypothesis that the implementation of agent-based artificial intelligence systems leads to increased efficiency and quality of decision-making and management processes, provided that it is accompanied by appropriate mechanisms for supervision, cybersecurity, and ethical control. The autonomy of AI agents, supported by long-term memory and the ability to monitor the state of the environment, significantly increases the effectiveness of multi-stage planning and resource management in conditions of volatility and uncertainty, which translates into better allocation of capital, time, and information in organizations. At the same time, the level of operational and economic risk remains strongly correlated with the degree of integration of technical safeguards, decision interpretability, and audit procedures, as the absence of these elements can lead to an escalation of autonomous errors and financial losses. The analysis also indicates that the implementation of cybersecurity standards, such as continuous monitoring of agents' activities and segmentation of data access, significantly reduces the vulnerability of systems to attacks and manipulation. With regard to the third research question, it was found that ethical and regulatory conditions act as a moderating factor in the pace and scale of agent AI adoption, increasing stakeholder trust and social acceptance of the technology. Organizations operating in a clear regulatory environment are more likely to opt for broader implementations, treating agentic AI as a tool that supports rather than replaces the responsibility of decision-makers. The analytical commentary indicates that the synergistic combination of technological autonomy with institutional and ethical frameworks determines the long-term usefulness of agentic AI as a safe and effective element of a modern knowledge-based economy.

Agentic artificial intelligence represents a new paradigm in the development of digital technologies. While earlier waves of innovation focused on creating static tools, we are now seeing the emergence of systems capable of making decisions, conducting complex social interactions, and collaborating in multi-agent environments (Cross et al. 2024). The impact of this change is interdisciplinary, affecting software engineering and robotics as well as social sciences, cognitive science, and ethics. The basic pillar of agent-based AI development is the concept of agent architectures, the most well-known of which remains the BDI model – *Belief, Desire, Intention*. Developed in the early 1990s, this model continues to inspire the design of autonomous systems, enabling the implementation of behaviors close to human decision-making processes (de Silva and Meneguzzi 2020).

In recent years, there has been growing interest in the formal verification of BDI systems. Thanks to the use of formal logic tools, it is possible to test the correctness of agents' actions, which is important in critical applications such as nuclear safety and medical systems (Aarhus University, n.d.). The example of verifying a nuclear inspection robot shows that agent-based systems can detect and correct errors in high-risk environments. At the same time, multi-agent systems are being developed, in which coordination, negotiation, and adaptability play a key role. In this context, the ability of AI to develop Theory of Mind (ToM) – understanding the mental states of other agents and people – is particularly important.

The ability to model the mental states of others is crucial for agents operating in social environments. Research by Graziano and Becchio (2024) shows that models such as GPT-4 can achieve results in Theory of Mind tests that are comparable to, and sometimes superior to, those of humans. This is particularly true for tasks related to understanding false beliefs or interpreting indirect requests.

However, according to Nguyen (2025), these abilities are limited and unstable - models are sensitive to context and may “simulate” ToM rather than actually understand intentions. The imperfection of ToM benchmarks also remains a problem. Riemer et al.

(2024) point out that most tests only examine “literal ToM” rather than “functional ToM,” i.e., the ability to adapt consistently during longer interactions.

New architectures are proposed, such as “Hypothetical Minds” (Cross et al. 2024), which embed mechanisms for generating and testing hypotheses about the behavior of other agents into the agent. This approach significantly improves performance in multi-agent tasks, suggesting a future direction for development.

The growing capabilities of agentic AI come with challenges in terms of security and ethics. Bai et al. (2022) proposed a *Constitutional AI* approach, in which systems learn ethical behavior through an internal “constitution” - a set of rules that guide them during interactions. This solution reduces the need for manual data labeling by humans, while increasing the consistency and safety of responses.

However, Nguyen (2023) points out that the implementation of Constitutional AI raises new questions: who defines the constitution, how can it be adapted to different cultures and legal systems, and does this approach lead to excessive “normativization” of AI? This field of research opens up an intense interdisciplinary debate at the intersection of social sciences, law, and computer science.

In view of the above, the development of agentic artificial intelligence in 2024–2025 clearly shows a paradigm shift from tool-based systems to interactive, autonomous, and adaptive agents that can operate in complex social and technical environments. This phenomenon results both from advances in large language model (LLM) technology and from the growing computing power of data centers and the ability to integrate diverse agents into multi-agent systems with increasingly general competencies (Cross et al., 2024; de Silva and Meneguzzi, 2020). These systems, including AI data centers, Industry 4.0 technologies implemented in business entities, including solutions based on generative artificial intelligence and computerized Big Data Analytics platforms (Kwasek, Prokopowicz 2023), are an important element of the economic development of a country based, among other things, on the development of technological industries (Pająk, Prokopowicz, Rawska, Soroka, Karpiński 2024).

An analysis of agent architectures, including classic BDI systems and their contemporary extensions, indicates that the foundation for building autonomous agents remains the ability to represent beliefs, goals, and intentions, and to make decisions in a manner similar to human reasoning (de Silva and Meneguzzi, 2020). These systems not only enable independent operation in dynamic environments, but also integration with other agents, creating networks of cooperating entities that can adapt their strategies in real time in response to changing conditions (Aarhus University, n.d.; Riemer et al., 2024). In this context, the development of tools for formal verification of the correctness of agents' behavior is crucial, especially in critical applications such as medicine, energy, or nuclear systems, where an error can have catastrophic consequences (Bai et al., 2022).

The ability of agents to model the mental states of other participants in an interaction, known as Theory of Mind (ToM), remains a central element of cooperation and coordination in multi-agent systems. Research shows that modern LLMs, such as GPT-4, can achieve high scores on ToM tasks, including understanding false beliefs, interpreting subtle cues, and predicting the actions of other agents (Graziano and Becchio, 2024; Kosinski, 2024). At the same time, there are limitations related to the sensitivity of these models to context and the risk of superficial simulation of mental states rather than their full understanding (Nguyen, 2025). The limitations of ToM benchmarks, including the lack of assessment of the ability to adapt consistently in long-term interactions, highlight the need for further development of architectures such as “Hypothetical Minds,” which incorporate mechanisms for generating and testing hypotheses about the behavior of other agents, increasing effectiveness in multi-agent environments (Cross et al., 2024).

The safety and ethics of agentic AI require a new approach that integrates both moral norms and technical mechanisms. Constitutional AI, proposed by Bai et al. (2022), introduces the concept of an internal “constitution,” a set of ethical principles that guide an agent's decision-making. This approach reduces the need for manual data labeling and increases the consistency of responses. However, Nguyen (2023) emphasizes that defining a constitution raises questions about the universality of norms, their adaptation to different cultures and legal systems, and the risk of excessive normalization, which requires an interdisciplinary debate combining computer science, law, social sciences, and philosophy.

The outlook for the coming years suggests that the development of agentic AI will involve not only improving individual agents in terms of autonomy, adaptability, and understanding of social contexts, but also integrating different types of agents into a so-called “super-AI” with general functionality (general AI). Such systems could combine the competencies of highly specialized agents capable of cooperation, negotiation, and decision-making in many fields, opening up new possibilities in areas such as management, scientific research, environmental protection, and medicine (Bai et al., 2022; Cross et al., 2024; Riemer et al., 2024).

The conclusions drawn from the interdisciplinary analysis point to the need for continuous monitoring of the effects of introducing autonomous agents, including their impact on employment structures, social relations, legal norms, and the security of critical systems. The implementation of AI agents requires the development of regulatory frameworks, audit mechanisms, and verification tools to ensure that the systems' operations comply with ethical and safety principles (Brundage et al., 2023; Nguyen, 2023).

In view of the above, the main conclusions from the analysis are as follows:

1. Agent architectures (BDI and their extensions) provide the foundation for designing agents capable of reasoning and acting in complex environments.
2. Theory of Mind in LLM is crucial for cooperation and coordination in multi-agent systems, although current models still struggle with limitations and the risk of superficial interpretation.
3. Ethics and security require new approaches - such as *Constitutional AI* - which, however, open up a broader discussion about the norms and culture governing the development of technology.

In summary, the years 2024–2025 point to a transformation of the AI paradigm – from tool-based systems to autonomous agents capable of complex cooperation, understanding mental states, and acting in accordance with accepted norms. This development opens up both scientific and practical perspectives, while requiring consideration of technical, ethical, legal, and social aspects to maximize the benefits and minimize the risks associated with the implementation of agentic AI. In the coming years, we can expect not only the improvement of individual AI agents, but also the emergence of systems integrating different types of agents into so-called “super-AI” with general functionality.

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